

3. Evaluation Metrics

3.1 Confusion Matrix

A confusion matrix summarizes the performance of a classification model by tabulating the counts of true positives, false positives, true negatives, and false negatives. It is useful for understanding the types of errors made by the model and assessing its performance across different classes.

The confusion matrix is typically organized with the following side headings:

- **True Positives (TP):** Instances that are actually positive and are correctly predicted as positive.
- **False Positives (FP):** Instances that are actually negative but are incorrectly predicted as positive.
- **True Negatives (TN):** Instances that are actually negative and are correctly predicted as negative.
- **False Negatives (FN):** Instances that are actually positive but are incorrectly predicted as negative.

ACTUAL ↓	PREDICTED	
	POSITIVE	NEGATIVE
POSITIVE	TRUE POSITIVE (TP)	FALSE POSITIVE (FP)
NEGATIVE	FALSE NEGATIVE (FN)	TRUE NEGATIVE (TN)

3.2 Accuracy

Accuracy is the most basic evaluation metric, representing the proportion of correct predictions out of the total predictions made. It is calculated as the ratio of the number of correctly classified instances to the total number of instances. However, accuracy alone may not provide a complete picture, especially when the classes are imbalanced.

3.3 Precision

Precision measures the proportion of correctly predicted positive instances out of all predicted positive instances. It focuses on the accuracy of positive predictions. Precision is calculated as the ratio of true positives (correctly predicted positives) to the sum of true positives and false positives (incorrectly predicted positives).

3.4 Recall or Sensitivity or True Positive Rate

Recall measures the proportion of correctly predicted positive instances out of all actual positive instances. It focuses on the model's ability to find all positive instances. Recall is calculated as the ratio of true positives to the sum of true positives and false negatives (missed positives).

3.5 F1 Score

The F1 score combines precision and recall into a single metric, providing a balanced evaluation. It is the harmonic mean of precision and recall, calculated as $2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$. The F1 score is useful when the class distribution is imbalanced.

3.6 Specificity or True Negative Rate

Specificity measures the proportion of correctly predicted negative instances out of all actual negative instances. It focuses on the model's ability to identify negative instances. Specificity is calculated as the ratio of true negatives (correctly predicted negatives) to the sum of true negatives and false positives (incorrectly predicted negatives).
Specificity (True Negative Rate)

3.7 Formulae for the above discussed metrics

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN} \quad \text{Precision} = \frac{TP}{TP+FP} \quad \text{Recall (Sensitivity)} = \frac{TP}{TP+FN}$$

$$\text{F1 Score} = \frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad \text{Specificity (True Negative Rate)} = \frac{TN}{TN+FP}$$

3.8 Area Under the Receiver Operating Characteristic Curve (AUC-ROC)

The ROC curve is a graphical representation of the model's performance at different classification thresholds. AUC-ROC represents the area under the ROC curve, which provides an aggregate measure of the model's discrimination ability. A higher AUC-ROC indicates better classification performance.

To find,

1. Sort the predicted probabilities or scores in descending order.
2. Calculate the True Positive Rate (TPR) and False Positive Rate (FPR) for each threshold. TPR is the ratio of true positives to the total number of actual positives, and FPR is the ratio of false positives to the total number of actual negatives.
3. Plot the points on a graph with TPR on the y-axis and FPR on the x-axis.
4. Calculate the area under the curve using a suitable numerical integration method, such as the trapezoidal rule or Simpson's rule.

